Week 10, Lecture 1: Likelihood Ratio Tests, AIC, and Regularization

Instructor notes. Part 1: prep reading. Part 2: podium cheat sheet

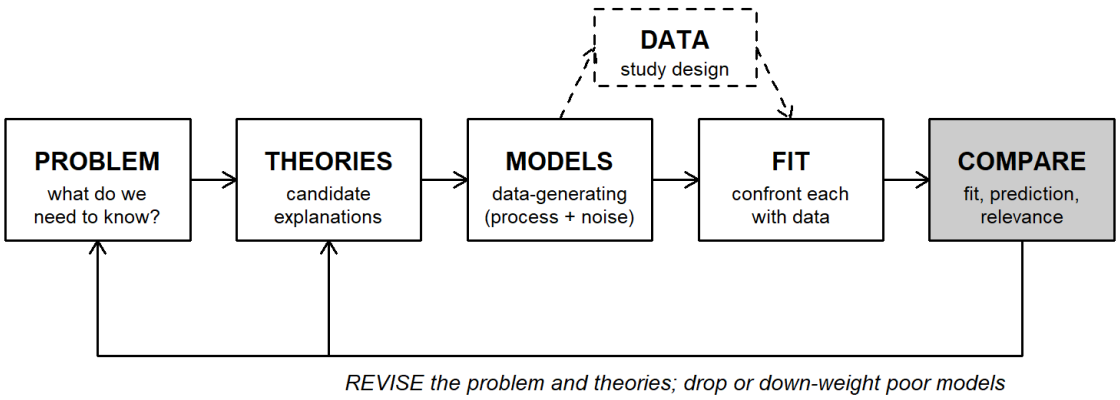
Overview (instructor only)

Beat	Mode	Time
1. Big picture	BOARD	~4 min
2. WOD: *Connect the Dots*	handout	~8 min
3. Likelihood ratio test	BOARD → R	~8 min
4. AIC	BOARD → R	~10 min
5. Ridge and lasso	BOARD → R	~12 min
6. Wrap and discussion	BOARD	~5 min

- R = Monday section of `CODE_Week10_ModelComparison.R`. No Stan debrief (lab).
- **Behind?** Skip the Poisson pause and AICc/BIC; past minute ~38, do section 5's board part only and open Wednesday with the glmnet demo.

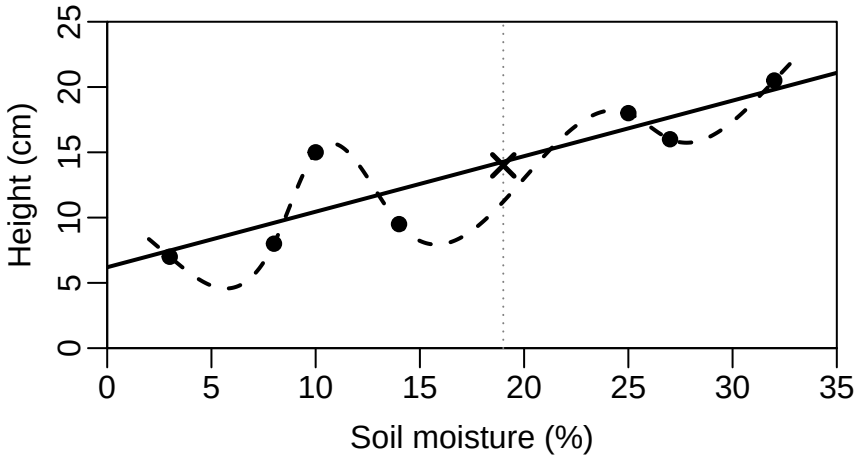
Part 1: Lecture content

1. Big picture (~4 min) [BOARD]



- COMPARE = fit / prediction and rivals / relevance; today is the middle one
- Tools: penalize in-sample fit (LRT, AIC), held-out data (CV), or fit skeptically (ridge, lasso)
- Survivors are corroborated, not proven

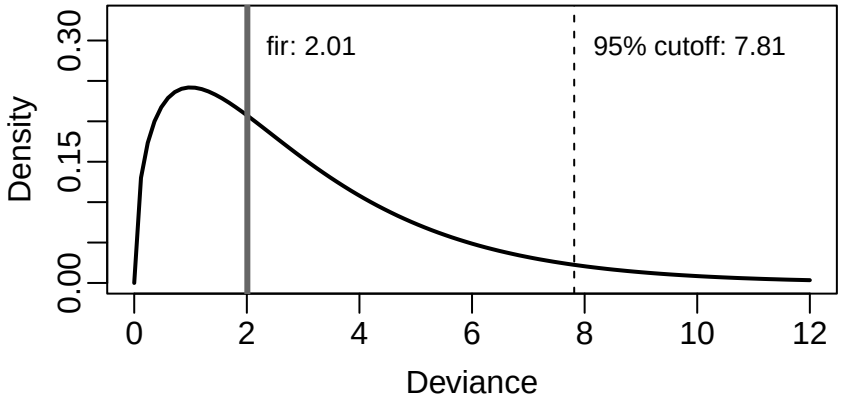
2. WOD reveal (~8 min) [handout]



- Residual SS: line 35.2, curve 0
- New plot **14 cm**; line 14.3, curve 11.2
- Land: in-sample fit always improves with flexibility

3. Likelihood ratio test (~8 min) [BOARD → R]

- Fir (Bolker 6.6): cones $\sim$ NegBin$(a\,\mathrm{DBH}^b, k)$; full (wave/non-wave a, b, k; 6 params) vs. no-wave (3)
- $D = 2(\mathrm{NLL}_{red} - \mathrm{NLL}_{full}) \sim \chi^2_r$; nested models only
- Week 7's 1.92 = this test with $r = 1$



- **Result:** $D = 2.01$, $p = 0.57$; keep no-wave

- **Pause:** Poisson nested in NB? Yes ($k \to \infty$; legacy LECTURE8 is wrong), but on the boundary, so naive p is ~2x too big. $D = 4{,}052$ here
- **Sticking point:** not rejecting $\neq$ no difference

4. AIC (~10 min) [BOARD → R]

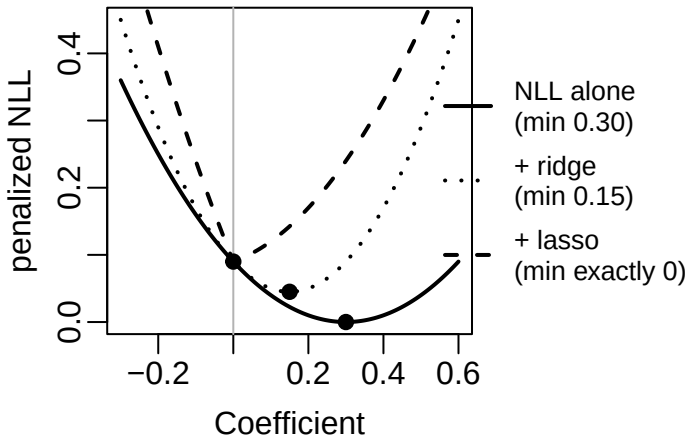
- AIC $= -2 \log \mathcal{L}(\hat{\theta}) + 2k$; nested or not, same response data
- $2k$ = average in-sample optimism (`NOTES_Reference_AICPenalty.Rmd`; don't derive)
- $w_i = e^{-\Delta_i/2} / \sum_j e^{-\Delta_j/2}$; BIC: $k \log n$

	k	NLL	AIC	dAIC	weight	dBIC
no_wave	3	1136.02	2278.03	0.00	0.52	0.00
const_ak	4	1135.76	2279.52	1.48	0.25	4.97
const_b	5	1135.13	2280.27	2.24	0.17	9.21
full	6	1135.01	2282.02	3.99	0.07	14.46
poisson	2	3161.83	6327.66	4049.63	0.00	4046.14

- **Pause:** weight 0.52 = P(true model)? No; relative support within the set (Bolker p. 284)
- **Sticking point:** `const_ak` adds 1 param, NLL down only 0.26, yet $\Delta = 1.48$. A useless parameter costs ≤ 2 (Arnold 2010)

5. Ridge and lasso (~12 min) [BOARD → R]

- Creepers: 50 stations, 10 covariates, simulated (say so); truth = big trees, snags, road distance
- $2^{10} = 1024$ subsets; instead penalize coefficient *size*: NLL$+\lambda \sum \beta_j^2$ (ridge), NLL$+\lambda \sum |\beta_j|$ (lasso)
- Standardize first; λ by 10-fold CV (`lambda.1se`)
- Lasso's corner at 0 gives exact zeros:



- **Full MLE:** all 10 nonzero; north 0.22, elev 0.11 (truth 0)
- **All-subsets AIC:** big_trees, snags, north, road_dist; 8 within 2; best weight 0.08
- **Top-models table:** the true model ranks 2 (ΔAIC 0.86). 6 of the next 9 are the best model plus one covariate, at ΔAIC 1.72 to 1.99: uninformative parameters (section 4, Arnold 2010). Pause: which covariates are in nearly every top model?
- **Lasso (CV):** big_trees, snags, road_dist; big trees 0.37 (truth 0.5), road 0.07 (0.3)
- **Ridge (CV, same folds and rule):** no zeros, everything shrunk hard: big trees 0.20, snags 0.22, road 0.08, north 0.02. At `lambda.min` much less: big trees 0.51, road 0.29, north 0.13 (mention only if asked)
- **40 simulated datasets**, exact true set: AIC 12, lasso 13
- **Pause:** road 0.07 vs. 0.3, broken? No: bias traded for lower variance; for effect sizes refit or go Bayesian (Wed.)
- **Sticking point:** neither method is a truth detector

6. Wrap and discussion (~5 min) [BOARD]

- Count penalty / held-out data / size penalty. Tomorrow: pond survey. Wed.: WAIC, LOO, shrinkage priors

Q1. AIC kept northness (8 models within 2, best weight 0.08); the lasso dropped it. Which covariates would you report as "important," and how would you word it?

Q2. In a real survey, the lasso sets grazing to exactly 0. A manager concludes expanding grazing won't affect creepers. Agree?

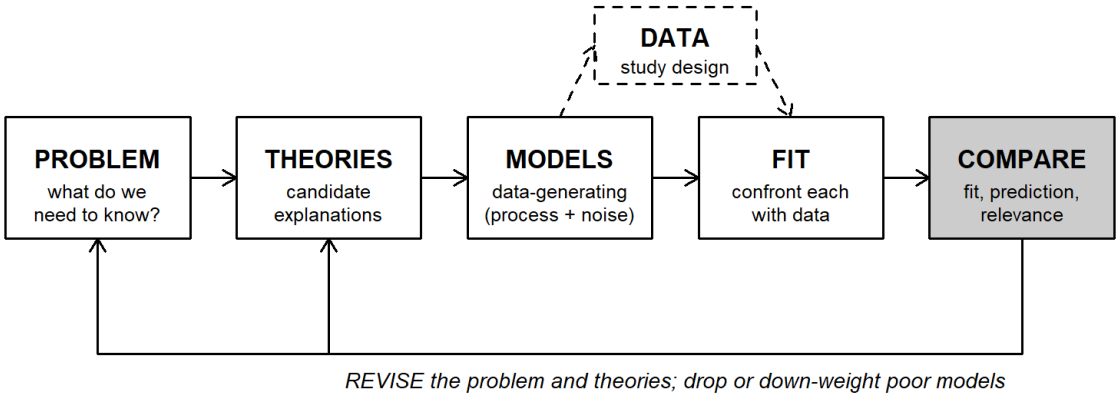
- Q1 lands on: report selection uncertainty; agreed covariates (big trees, snags) are safest
- Q2 lands on: zero = not needed for prediction here, not "no effect"; grazing may act via shrub cover; causal question = COMPARE (3), relevance

PART 2: PODIUM CHEAT SHEET

Tight on purpose. Detail: Part 1. R = Monday section of CODE_Week10_ModelComparison.R.

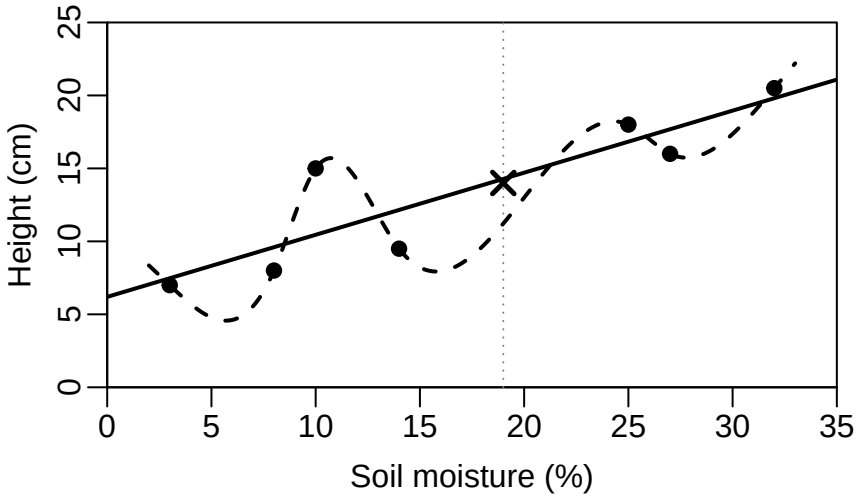
Order: big picture → WOD → lecture (no Stan debrief; that's at lab). **Behind?** Skip the Poisson pause and the AICc/BIC columns; move the glmnet demo to Wednesday.

1. Big picture (~4 min) [BOARD]



- Point at **COMPARE**: (1) does it fit, (2) does it beat its rivals and predict new data, (3) is it relevant to the problem
- Weeks 6-9 were FIT; today is (2): which models do the data support?
- Three routes: penalize in-sample fit (LRT, AIC); test on new data (CV); fit one big model skeptically (ridge, lasso)
- Survivors are corroborated, not proven
- Hand out the WOD

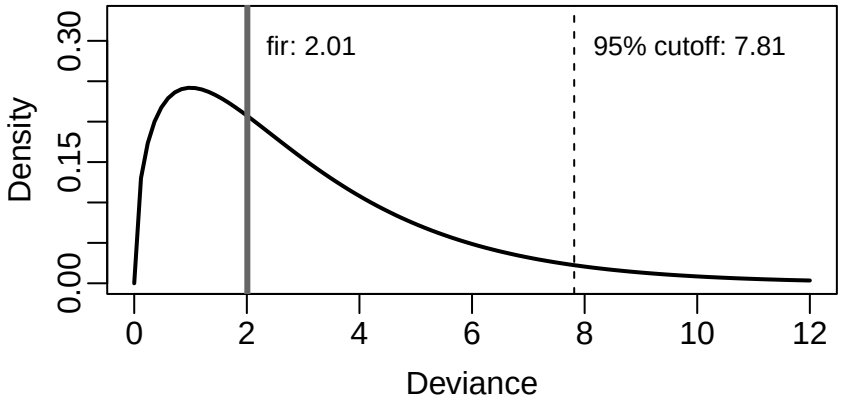
2. WOD reveal (~8 min)



- Sum of squared residuals: line 35.2, curve 0. The curve "wins" on the data it was fit to
- New plot measured **14 cm**. Line predicted 14.3; this curve 11.2
- Compare neighbors: curves disagree in the gap, lines don't
- **Land:** in-sample fit always improves as a model gets more flexible, so we need a penalty or new data

3. Likelihood ratio test (~8 min) [BOARD → R]

- Fir cones ~ NegBin(mean $a\,\mathrm{DBH}^b$, dispersion k). Do wave sites differ?
- Full: separate a, b, k for wave and non-wave (6 params). Reduced: one set (3). Reduced is a special case of full: **nested**
- Deviance $D = 2(\mathrm{NLL}_{red} - \mathrm{NLL}_{full}) \sim \chi^2_r$, r = params fixed
- Callback: Week 7's 1.92 cutoff = this test with $r = 1$ (3.84/2)



- **R:** $D = 2.01$ vs. cutoff 7.81 ($p = 0.57$): keep the simpler no-wave model
- **Pause (skippable):** "Is the Poisson model nested in the negative binomial? Can we use an LRT?" *Land:* yes, nested ($k \to \infty$), but $k = \infty$ is on the boundary, so the naive p is about twice too big. Here $D =$ 4,052, so it doesn't matter
- **Sticking point:** not rejecting ≠ "wave and non-wave trees are the same"; 242 trees just can't tell them apart

4. AIC (~10 min) [BOARD → R]

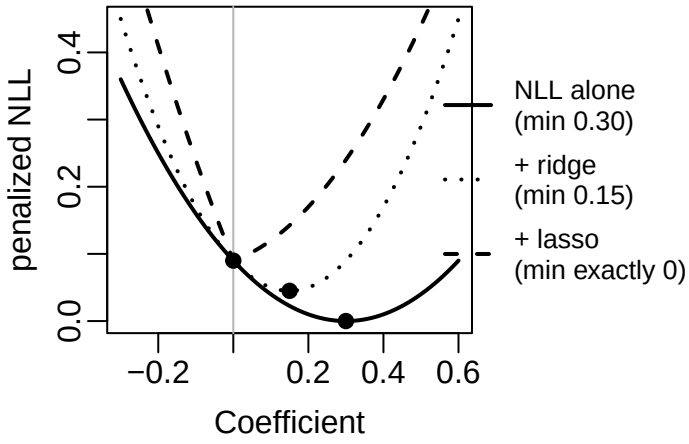
- AIC $= -2\log\mathcal{L}(\hat{\theta}) + 2k$: ranks any models fit to the same response, nested or not
- $2k$ = average optimism of in-sample fit (the WOD's curve, in one number); don't derive
- $\Delta_i = \text{AIC}_i - \text{AIC}_{min}$; weights $w_i = e^{-\Delta_i/2} / \sum_j e^{-\Delta_j/2}$
- AICc: small-sample correction. BIC: $k \log n$ in place of $2k$ (harsher)

	k	NLL	AIC	dAIC	weight	dBIC
no_wave	3	1136.02	2278.03	0.00	0.52	0.00
const_ak	4	1135.76	2279.52	1.48	0.25	4.97
const_b	5	1135.13	2280.27	2.24	0.17	9.21
full	6	1135.01	2282.02	3.99	0.07	14.46
poisson	2	3161.83	6327.66	4049.63	0.00	4046.14

- **Pause:** "The no-wave model has weight 0.52. Is there a 52% chance it's the true model?" *Land:* no; relative support within this set, not a probability (Bolker p. 284)
- **Sticking point:** `const_ak` adds 1 param, NLL drops only 0.26, yet it's only 1.48 behind. A useless parameter costs at most 2, so "within 2" $\neq$ support (Arnold 2010)

5. Ridge and lasso (~12 min) [BOARD → R]

- Creepers: 50 point counts, 10 standardized habitat covariates. **Simulated, so we know the truth:** big trees, snags, road distance
- $2^{10} = 1024$ subsets. Instead, keep all 10 and penalize coefficient *size*: NLL + $\lambda \sum \beta_j^2$ (ridge), NLL + $\lambda \sum |\beta_j|$ (lasso)
- AIC penalizes how many; these penalize how big. Standardize first



- Lasso's corner at 0 gives exact zeros; ridge only shrinks
- **R, MLE:** all 10 nonzero; northness 0.22, elevation 0.11 (both truly 0)
- **R, all-subsets AIC:** best = big_trees, snags, north, road_dist (north spurious); 8 models within 2; top weight only 0.08
- **R, top-models table:** truth ranks 2 (ΔAIC 0.86); 6 of the next 9 = best model + one useless covariate, each at $\Delta \approx 2$. **Pause:** "Which covariates show up in nearly every top model?" *Land:* big trees, snags, road; the rest come and go
- **R, CV lasso** (`lambda.1se`): keeps big_trees, snags, road_dist, zeroes the rest
- **R, CV ridge** (same folds): no zeros; big trees 0.20, road 0.08
- **Pause:** "The lasso kept road distance but says its effect is 0.07; the truth is 0.3. Is the lasso broken?" *Land:* no. Shrinkage is the point: biased, less noisy estimates that predict better. Good for prediction and choosing covariates, not for effect sizes (refit, or Bayesian on Wednesday)
- Across 40 simulated datasets, exact true set: AIC 12, lasso 13. Neither is a truth detector

6. Wrap and discussion (~5 min)

- Three ways to compare: penalize the count (LRT, AIC), test on held-out data (CV), penalize the size (ridge, lasso)
- Tomorrow: the pond survey. Wednesday: Bayesian versions (WAIC, LOO, shrinkage priors)

Q1. AIC's best creeper model kept northness (8 models within 2 units; top weight 0.08); the lasso dropped it. Which covariates would you report as "important," and how would you word it?

- *Land:* report the uncertainty, not one list; covariates both methods agree on (big trees, snags) are the safest claims

Q2. In a real survey, the lasso sets grazing to exactly 0. A manager concludes expanding grazing won't affect creepers. Agree?

- *Land:* a zero means "not needed to predict counts here," not "no effect"
- Grazing may act through shrub cover, or be real but too small for 50 stations. A management question is causal: COMPARE (3), relevance